Prime Factorization Escape Folder

Unit 1: Number System Launch · Lesson 1-1 · Flagship · EduWonderLab

UNIT 1 · LESSON 1-1 ★ FLAGSHIP · TEACHER NOTES — **Standard 6.NS.4**

FORMAT **Cooperative escape challenge**	TIME **30–35 min**
GROUPING **Teams of 3–4**	TOPIC **Prime Factorization (Flagship)**
STANDARD **6.NS.4**	PREP LEVEL **Medium — print clue cards into 4 envelopes per team**

Learning goal *I can use prime factorization to solve a chain of clues and crack a final code.*

Teacher setup

- Print one clue set per team. Place each clue in its own envelope labeled 1–4.
- Teams open one envelope at a time. Each prime factorization reveals one code digit.
- Hold the final-lock card until a team thinks they have all four digits.

Materials

4 envelope clue cards, Factor-tree scratch paper, Code strip, Pencil.

Differentiation & language support

- Provide a worked factor tree for 12 as a model on the table (ESOL/SPED).
- Pair a Recorder with a Checker so every digit is verified before moving on.
- Sentence stem: "The exponent on ___ is ___, so the code digit is ___."

Key vocabulary (English / Español):

Term	Español	What it means
exponent	exponente	tells how many times a base is multiplied
base	base	the number being multiplied
prime factorization	factorización prima	a product of only prime numbers

Quick teacher check: *Check the code strip (3-2-2-2). If a team is stuck, point them to the prime they missed.*

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Name: **Date:** **Class:**

Your goal: *I can use prime factorization to solve a chain of clues and crack a final code.*

What you need

Clue cards, Factor-tree scratch paper, Code strip.

How to play

1. Open Envelope 1. Find the prime factorization, then read the digit clue.
2. Write the code digit on your code strip. Move to the next envelope.
3. After Envelope 4, enter all four digits in order on the final lock card.
4. Finish with the challenge card to prove your team understands factor trees.

Clue Envelopes

Each clue gives one code digit. Find all four digits in order, then open the final lock.

1. Envelope 1: Find the prime factorization of 40. Code digit 1 = the exponent on 2.
2. Envelope 2: Find the prime factorization of 18. Code digit 2 = the exponent on 3.
3. Envelope 3: Find the prime factorization of 50. Code digit 3 = the exponent on 5.
4. Envelope 4: Find the prime factorization of 245. Code digit 4 = the exponent on 7.
5. Final Lock: Enter your four code digits in order to open the capsule.
6. Challenge Card: Build two different factor trees for 72 and write the prime factorization.

Code strip (write each digit as you solve):

Explain your thinking

Explain how an exponent in a prime factorization is different from a regular factor.

Sentence frame: *The exponent ___ means the prime ___ is used ___ times.*

★ **Challenge:** *Create your own clue for a teammate: choose a number, write its prime factorization, and name which exponent is the secret digit.*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. Envelope 1: Find the prime factorization of 40. Code digit 1 = the exponent on 2.
Answer: **$40 = 2^3 \times 5$ → digit 1 = 3**

2. Envelope 2: Find the prime factorization of 18. Code digit 2 = the exponent on 3.
Answer: **$18 = 2 \times 3^2$ → digit 2 = 2**

3. Envelope 3: Find the prime factorization of 50. Code digit 3 = the exponent on 5.
Answer: **$50 = 2 \times 5^2$ → digit 3 = 2**

4. Envelope 4: Find the prime factorization of 245. Code digit 4 = the exponent on 7.
Answer: **$245 = 5 \times 7^2$ → digit 4 = 2**

5. Final Lock: Enter your four code digits in order to open the capsule.
Answer: **3 – 2 – 2 – 2**

6. Challenge Card: Build two different factor trees for 72 and write the prime factorization.
Answer: **Both trees give $72 = 2^3 \times 3^2$**

Teaching notes & look-fors

- Final code is 3-2-2-2.
- Teams often forget the lone prime (the 5 in 40, the 5 in 245). Remind them every branch must end prime.